

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester M.C.A. (CBCS) Degree Examinations
 December 2024

22MCA352 – NATURAL LANGUAGE PROCESSING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is Natural Language Processing? What are the components of NLP and explain its significance in modern technology.	10	L*1	1	1
b)	What are the applications of NLP? Explain with examples.	10	L2	1	1
2. a)	What is Linguistics? What are the different types of Linguists?	10	L2	1	1
b)	How does the flow of Information take place in Natural Language Systems? Explain with a diagram.	10	L2	1	1
Unit – II					
3. a)	What is Text Normalization? Explain the different steps in text normalization using examples.	10	L1	2	1
b)	What is Minimum Edit Distance? List some applications. Calculate the minimum edit distance required to convert the word "Hospital" to "Hotspot," using Dynamic Programming and Backtracking. List the operations involved.	10	L3	2	1
4. a)	Describe a Hidden Markov Model (HMM) and outline the key assumptions used to determine the probability of being in a particular state at a specific time?"	10	L2	2	1
b)	Solve the following using Markov chains: In a certain market, only two brands of cold drinks A & B are sold. Given that a man last purchased brand A, there is 80% chance that he would buy the same brand in the next purchase, while if a man purchased brand B, there is a 90% chance that his next purchase would be brand B. Using this information: i. Develop a transition probability matrix. ii. Develop a transition probability Diagram iii. Interpret the state transition matrix in terms of retention & loss and retention & gain	10	L3	2	1
Unit – III					
5. a)	Perform a top-down chart parsing for the following statement: "The large can can hold the water" Grammar Rules: S → NP VP NP → ART ADJ N ART N ADJ N VP → AUX VP V NP Lexicon Entries: the: ART, large: ADJ, can: N,AUX,V, hold: N,V, water: N, V	10	L3	3	2
b)	Explain the bottom-up chart parsing method and highlight its advantages compared to other parsing techniques?"	10	L2	3	1

P.T.O.

6. a) Perform a bottom-up tree parsing for the following statement:
"The small tree shades the new house by the stream"

Grammar Rules:

$S \rightarrow NP VP$

$NP \rightarrow Det Adj N$

$NP \rightarrow Det N$

$PP \rightarrow P NP$

$VP \rightarrow V NP PP$

10 L3 3 2

- b) What are the characteristics of Depth First Strategy and Breadth First Strategy? Mention any 5 key differences between DFS and BFS

10 L2 3 1

Unit – IV

7. a) Define feature systems and augmented grammars in the context of syntactic analysis. How do they enhance the representation of linguistic information beyond traditional grammars?

10 L2 4 1

- b) Define binary features and provide two examples of binary features used in linguistic analysis.

10 L2 4 1

8. a) Describe the process of morphological analysis and its importance in understanding word formation. Provide an example of morphological analysis for the word "unbelievably".

10 L2 4 1

- b) Explain the role of the lexicon in language processing and its relationship with feature systems. How does the lexicon contribute to morphological analysis?

10 L2 4 1

Unit – V

9. a) Discuss the methods used for temporal processing in text. How can these methods be used to extract and interpret time-related information from a document?

10 L2 5 1

- b) Explain how relation detection and classification work in the context of information extraction. Illustrate with an example of a detected relationship between entities in a sentence.

10 L2 5 1

10. a) What is template filling in the context of information extraction? Explain how template filling is used to structure information extracted from text and provide examples of its application in real-world scenarios.

10 L2 5 1

- b) What are the key techniques used in event processing for information extraction? Provide an example of how event processing can be applied to identify and analyze events in a text.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
